



Domination number, independent domination number and k -independence number in trees

Qing Cui^{*}, Xu Zou

School of Mathematics, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, PR China

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ABSTRACT

For any graph G , let $\gamma(G)$ and $i(G)$ denote the domination number and the independent domination number of G , respectively. For any positive integer k , a subset S of vertices in a graph G is said to be a k -independent set of G if $G[S]$ has maximum degree less than k . The k -independence number of G , denoted by $\alpha_k(G)$, is the maximum cardinality of a k -independent set of G . Let T be any tree with $n \geq 2$ vertices. Dehgardi et al. proved that $i(T) \leq \frac{3}{4}\alpha_2(T)$ and $\gamma(T) + i(T) \leq \frac{4}{3}\alpha_2(T)$. Later, Zhang and Wu extended the former result of Dehgardi et al. by showing that $i(T) \leq \frac{k+1}{2k}\alpha_k(T)$, and conjectured that the latter one can also be generalized to $\gamma(T) + i(T) \leq \frac{2k}{2k-1}\alpha_k(T)$. In this paper, we prove this conjecture, and moreover, we characterize all extremal trees for which the equality holds.

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1. Introduction

In this paper we only consider finite simple graphs. For a graph G , we use $V(G)$ and $E(G)$ to denote the vertex set and the edge set of G , respectively. For any $v \in V(G)$, the *degree* $d_G(v)$ (or just $d(v)$ if it is clear from the context) of v is the number of edges incident to v in G . The maximum degree of G is denoted by $\Delta(G)$. For any $S \subseteq V(G)$, we use $G[S]$ and $G - S$ to denote the subgraphs of G induced by S and $V(G) \setminus S$, respectively, and write $G - s$ instead of $G - \{s\}$ if $S = \{s\}$. For any $H \subseteq G$, we simply write $G - H$ instead of $G - V(H)$. The *diameter* of G , denoted by $\text{diam}(G)$, is the maximum distance between any pair of vertices of G . A *diametrical path* of G is a path whose length is equal to the diameter of G . Let K_n , P_n and $K_{1,n-1}$ denote the complete graph, the path and the star with n vertices, respectively. We write $A := B$ to rename B as A .

Let G be a graph. A subset $D \subseteq V(G)$ is a *dominating set* of G if every vertex in $V(G) \setminus D$ has at least one neighbor in D . The *domination number* of G , denoted by $\gamma(G)$, is the minimum cardinality of a dominating set of G . A subset $S \subseteq V(G)$ is an *independent set* of G if no two vertices in S are adjacent. The *independence number* of G , denoted by $\alpha(G)$, is the maximum cardinality of an independent set of G . A subset $I \subseteq V(G)$ is an *independent dominating set* of G if I is both independent and dominating. The *independent domination number* of G , denoted by $i(G)$, is the minimum cardinality of an independent domination set of G . Clearly, $\gamma(G) \leq i(G) \leq \alpha(G)$. The study of domination and its variations in graphs is one of the most popular subjects in graph theory and has been investigated extensively in the literature (see e.g., [1,4,8,9,12,18,19,21–23,25,26]).

In 1985, Fink and Jacobson [16,17] introduced the following notions which generalize the aforementioned concepts. Let k and j be two positive integers. A subset $D \subseteq V(G)$ is a *k -dominating set* of G if every vertex in $V(G) \setminus D$ has at least k neighbors in D . The *k -domination number* of G , denoted by $\gamma_k(G)$, is the minimum cardinality of a k -dominating set of

^{*} Corresponding author.

E-mail addresses: cui@nuaa.edu.cn (Q. Cui), zouxu_math@nuaa.edu.cn (X. Zou).

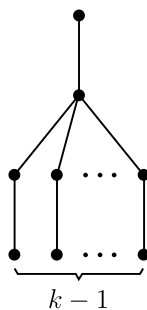


Fig. 1. The tree $S_{k,k-1}$.

G . A subset $S \subseteq V(G)$ is a k -independent set of G if $\Delta(G[S]) < k$. The k -independence number of G , denoted by $\alpha_k(G)$, is the maximum cardinality of a k -independent set of G . A subset $I \subseteq V(G)$ is a k -independent j -dominating set of G if I is both k -independent and j -dominating. The k -independent j -domination number of G , denoted by $i_j^k(G)$, is the minimum cardinality of a k -independent j -domination set of G . It is clear that $\gamma_1(G) = \gamma(G)$, $\alpha_1(G) = \alpha(G)$ and $i_1^1(G) = i(G)$. We refer the readers to [3,6,13,15,20] for more information on k -domination and k -independence in graphs.

In this paper, we mainly consider the related problems of domination and independence in trees. Favaron [14] proved that $i(T) \leq \frac{n+n_1}{3}$ for any tree T with $n \geq 2$ vertices and with n_1 vertices of degree one. Ma and Chen [28] provided a characterization of n -vertex trees T with $i(T) = \frac{n}{2}$. Lemańska [27] proved that $\gamma(T) \geq \frac{n-n_1+2}{3}$ for any tree T with $n \geq 3$ vertices and with n_1 vertices of degree one. Blidia et al. [2] showed that $\alpha_k(T) \geq \frac{k}{k+1}n$ for any positive integer k and any n -vertex tree T , and characterized all extremal trees attaining the bound. More results can be found in [5,7,11,24,30].

In 2021, Dehgardi et al. [10] considered the relations among domination number, independent domination number and 2-independence number in trees, and derived the following two theorems.

Theorem 1.1 (Dehgardi et al. [10]). For any tree T with $n \geq 2$ vertices, $i(T) \leq \frac{3}{4}\alpha_2(T)$.

Theorem 1.2 (Dehgardi et al. [10]). For any tree T with $n \geq 2$ vertices, $\gamma(T) + i(T) \leq \frac{4}{3}\alpha_2(T)$.

As an extension of Theorem 1.1, Zhang and Wu [29] further proved the following result.

Theorem 1.3 (Zhang and Wu [29]). Let k be a positive integer. For any tree T with $n \geq 2$ vertices, $i(T) \leq \frac{k+1}{2k}\alpha_k(T)$.

At the end of their paper, Zhang and Wu [29] also proposed the following conjecture which, if true, would generalize Theorem 1.2.

Conjecture 1.4 (Zhang and Wu [29]). Let k be a positive integer. For any tree T with $n \geq 2$ vertices, $\gamma(T) + i(T) \leq \frac{2k}{2k-1}\alpha_k(T)$.

In this paper, we prove this conjecture, and moreover, we characterize all extremal trees attaining the equality. The proof will be given in Section 2.

We conclude this section with some notations and terminology. Let T be a tree. A *leaf* of T is a vertex of degree one, and a *support vertex* of T is a vertex adjacent to a leaf. In particular, if $T \cong K_2$, then we arbitrarily choose one vertex of K_2 as a leaf and the other as a support vertex. The set of leaves and the set of support vertices of T are denoted by $L(T)$ and $S(T)$, respectively. For any vertex v in a rooted tree T , the *maximal subtree* at v , denoted by T_v , is the subtree of T induced by v and all descendants of v . In addition, let L_v denote the set of leaves adjacent to a vertex $v \in V(T)$. For two positive integers r and s , a *double star* $DS_{r,s}$ is a tree with exactly two vertices that are not leaves, with one vertex adjacent to r leaves and the other adjacent to s leaves.

For the sake of brevity, a dominating set of T with minimum cardinality is called a $\gamma(T)$ -set and an independent dominating set of T with minimum cardinality is called an $i(T)$ -set. Similarly, for any positive integer k , a k -independent set of T with maximum cardinality is called an $\alpha_k(T)$ -set.

2. Proof of the main result

In this section, we prove Conjecture 1.4 and characterize all extremal trees for which the equality holds.

Fix a positive integer k and let $S_{k,k-1}$ be the tree obtained from a $K_{1,k}$ by subdividing $k-1$ of its edges once (see Fig. 1 for an illustration). We define a family of trees $\mathcal{T}_k$ recursively as follows:

- (i) $S_{k,k-1} \in \mathcal{T}_k$;
- (ii) if $T' \in \mathcal{T}_k$, then $T \in \mathcal{T}_k$, where T is the tree obtained by adding an edge between a support vertex of T' and a support vertex of the tree $S_{k,k-1}$ (vertex-disjoint from T').

An example of the trees in $\mathcal{T}_k$ is depicted in Fig. 2.

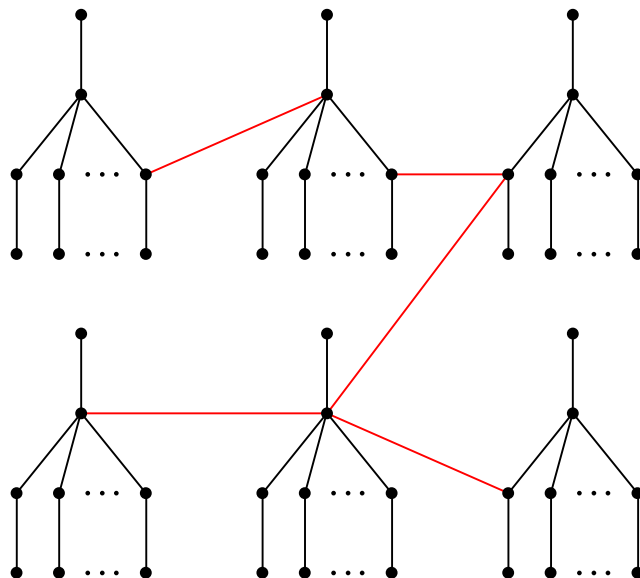


Fig. 2. An example of the trees in $\mathcal{T}_k$.

The following observation is easy to verify, hence we omit the proof here.

Observation 2.1. For any n -vertex tree $T \in \mathcal{T}_k$, the following statements hold:

- (a) $n \equiv 0 \pmod{2k}$;
- (b) every vertex is either a support vertex or a leaf;
- (c) every support vertex is adjacent to exactly one leaf;
- (d) $|L(T)| = |S(T)| = \frac{n}{2}$;
- (e) both $L(T)$ and $S(T)$ are $\gamma(T)$ -sets and hence $\gamma(T) = \frac{n}{2}$;
- (f) $L(T)$ is an $i(T)$ -set and hence $i(T) = \frac{n}{2}$;
- (g) $\alpha_k(T) = |L(T)| + \frac{k-1}{k}|S(T)| = \frac{2k-1}{2k}n$.

We are now ready to prove the main result of this paper.

Theorem 2.2. Let k be a positive integer. For any tree T with $n \geq 2$ vertices, $\gamma(T) + i(T) \leq \frac{2k}{2k-1}\alpha_k(T)$ with equality if and only if $T \in \mathcal{T}_k$.

Proof. If $T \in \mathcal{T}_k$, then it follows from Observation 2.1(e), (f) and (g) that

$$\gamma(T) + i(T) = \frac{n}{2} + \frac{n}{2} = n = \frac{2k}{2k-1} \cdot \frac{2k-1}{2k}n = \frac{2k}{2k-1}\alpha_k(T).$$

This proves the sufficiency of Theorem 2.2.

In the following, we will prove the necessity by induction on n . If $2 \leq n \leq 4$, then we see that $T \in \{K_2, P_3, P_4, K_{1,3}\}$ and it is easy to verify that the assertion of the theorem holds. So we may assume that $n \geq 5$.

If $\text{diam}(T) = 2$, then T is a star $K_{1,n-1}$. Since $\gamma(K_{1,n-1}) = i(K_{1,n-1}) = 1$ and $\alpha_k(K_{1,n-1}) \geq n-1$, we have

$$\gamma(K_{1,n-1}) + i(K_{1,n-1}) = 2 < n-1 < \frac{2k}{2k-1}(n-1) \leq \frac{2k}{2k-1}\alpha_k(K_{1,n-1}).$$

If $\text{diam}(T) = 3$, then T is a double star $DS_{r,s}$ with $r+s \geq 3$. Without loss of generality, we may assume that $r \geq s$ (and hence $r \geq 2$). If $r = 2$ and $s = 1$, then we conclude that $\gamma(DS_{2,1}) = i(DS_{2,1}) = 2$, $\alpha_k(DS_{2,1}) = 3$ when $k = 1$ and $\alpha_k(DS_{2,1}) \geq 4$ when $k \geq 2$, which implies that

$$\gamma(DS_{2,1}) + i(DS_{2,1}) = 4 < \frac{2k}{2k-1}\alpha_k(DS_{2,1}).$$

If $r = s = 2$, then we derive that $\gamma(DS_{2,2}) = 2$, $i(DS_{2,2}) = 3$, $\alpha_k(DS_{2,2}) = 4$ when $1 \leq k \leq 2$ and $\alpha_k(DS_{2,2}) \geq 5$ when $k \geq 3$, which means that

$$\gamma(DS_{2,2}) + i(DS_{2,2}) = 5 < \frac{2k}{2k-1} \alpha_k(DS_{2,2}).$$

If $r \geq 3$, then we know that $\gamma(DS_{r,s}) = 2$, $i(DS_{r,s}) = 1 + s$ and $\alpha_k(DS_{r,s}) \geq r + s$, and thus

$$\gamma(DS_{r,s}) + i(DS_{r,s}) = 3 + s \leq r + s < \frac{2k}{2k-1} (r + s) \leq \frac{2k}{2k-1} \alpha_k(DS_{r,s}).$$

Hence, we may assume that $\text{diam}(T) \geq 4$. Choose a diametrical path $v_1 v_2 \dots v_d v_{d+1}$ in T such that $d(v_2)$ is as large as possible, where $d := \text{diam}(T)$. Root T at v_{d+1} . We consider two cases according to the value of $d(v_2)$.

Case 1. $d(v_2) \geq 3$.

In this case, we have $|L_{v_2}| \geq 2$. We distinguish three subcases in terms of the value of k .

Subcase 1.1. $k = 1$.

Let $T' := T - v_1$, and let D' , I' and S' be a $\gamma(T')$ -set, an $i(T')$ -set and an $\alpha_1(T')$ -set, respectively. Define

$$D := \begin{cases} D', & \text{if } v_2 \in D', \\ (D' \setminus L_{v_2}) \cup \{v_2\}, & \text{if } v_2 \notin D', \end{cases}$$

$$I := \begin{cases} I', & \text{if } v_2 \in I', \\ I' \cup \{v_1\}, & \text{if } v_2 \notin I', \end{cases}$$

and

$$S := \begin{cases} (S' \setminus \{v_2\}) \cup L_{v_2}, & \text{if } v_2 \in S', \\ S' \cup \{v_1\}, & \text{if } v_2 \notin S'. \end{cases}$$

Then, it is straightforward to check that D is a dominating set of T implying that $\gamma(T) \leq |D| \leq \gamma(T')$ (where the second inequality holds since $L_{v_2} \setminus \{v_1\} \subseteq D'$ when $v_2 \notin D'$), I is an independent dominating set of T yielding that $i(T) \leq |I| \leq i(T') + 1$, and S is a 1-independent set of T implying that $\alpha_1(T) \geq |S| \geq \alpha_1(T') + 1$ (where the second inequality holds since $L_{v_2} \cap S' = \emptyset$ when $v_2 \in S'$). By the induction hypothesis, we deduce that

$$\gamma(T) + i(T) \leq \gamma(T') + i(T') + 1 \leq 2\alpha_1(T') + 1 \leq 2(\alpha_1(T) - 1) + 1 = 2\alpha_1(T) - 1 < 2\alpha_1(T).$$

Subcase 1.2. $k = 2$.

Let c_1, c_2, c_3 and c_4 be the numbers of children of v_3 with degree 1, 2, 3 and at least 4, respectively. Since $d(v_2) \geq 3$, we see that $c_3 + c_4 \geq 1$. Define $T' := T - T_{v_3}$. It is clear that any $\gamma(T')$ -set can be extended to a dominating set of T by adding v_3 and all its children with degree at least 2, which yields that $\gamma(T) \leq \gamma(T') + 1 + c_2 + c_3 + c_4$. Also, any $i(T')$ -set can be extended to an independent dominating set of T by adding all children of v_3 , which shows that $i(T) \leq i(T') + c_1 + c_2 + c_3 + c_4$. Moreover, any $\alpha_2(T')$ -set can be extended to a 2-independent set of T by adding all children of v_3 with degree at most 2 and all grandchildren of v_3 , which means that $\alpha_2(T) \geq \alpha_2(T') + c_1 + 2c_2 + 2c_3 + 3c_4$. By the induction hypothesis, we conclude that

$$\begin{aligned} \gamma(T) + i(T) &\leq \gamma(T') + i(T') + 1 + c_1 + 2c_2 + 2c_3 + 2c_4 \\ &\leq \frac{4}{3} \alpha_2(T') + 1 + c_1 + 2c_2 + 2c_3 + 2c_4 \\ &\leq \frac{4}{3} (\alpha_2(T) - c_1 - 2c_2 - 2c_3 - 3c_4) + 1 + c_1 + 2c_2 + 2c_3 + 2c_4 \\ &= \frac{4}{3} \alpha_2(T) + \frac{3 - c_1 - 2c_2 - 2c_3 - 6c_4}{3}. \end{aligned} \quad (1)$$

If $3 < c_1 + 2c_2 + 2c_3 + 6c_4$, then by (1), we have $\gamma(T) + i(T) < \frac{4}{3} \alpha_2(T)$. Therefore, we may assume that $3 \geq c_1 + 2c_2 + 2c_3 + 6c_4$. Since $c_3 + c_4 \geq 1$, we can derive that $c_3 = 1, c_2 = c_4 = 0$ and $0 \leq c_1 \leq 1$. This implies that $d(v_2) = 3$, and thus $|L_{v_2}| = 2$. Let $L_{v_2} = \{v_1, w\}$.

First, suppose $c_1 = 0$. Let D' , I' and S' be a $\gamma(T')$ -set, an $i(T')$ -set and an $\alpha_2(T')$ -set, respectively. Then, one can easily see that $D := D' \cup \{v_2\}$ is a dominating set of T implying that $\gamma(T) \leq |D| = \gamma(T') + 1$, $I := I' \cup \{v_2\}$ is an independent dominating set of T yielding that $i(T) \leq |I| = i(T') + 1$, and $S := S' \cup \{v_1, w\}$ is a 2-independent set of T implying that $\alpha_2(T) \geq |S| = \alpha_2(T') + 2$. By the induction hypothesis, we know that

$$\gamma(T) + i(T) \leq \gamma(T') + i(T') + 2 \leq \frac{4}{3} \alpha_2(T') + 2 \leq \frac{4}{3} (\alpha_2(T) - 2) + 2 = \frac{4}{3} \alpha_2(T) - \frac{2}{3} < \frac{4}{3} \alpha_2(T).$$

Next, suppose $c_1 = 1$. Then $|L_{v_3}| = 1$. Let z be the unique vertex in L_{v_3} . Substituting $c_1 = c_3 = 1$ and $c_2 = c_4 = 0$ into (1), we have $\gamma(T) + i(T) \leq \frac{4}{3} \alpha_2(T)$. Now, suppose $\gamma(T) + i(T) = \frac{4}{3} \alpha_2(T)$. Then all inequalities in (1) should be equalities, which shows that $\gamma(T') + i(T') = \frac{4}{3} \alpha_2(T')$. By the induction hypothesis, we deduce that $T' \in \mathcal{T}_2$ (and hence $|V(T')| \geq 4$). Then, it follows from Observation 2.1(b) that v_4 is either a support vertex or a leaf of T' .

- Suppose v_4 is a support vertex of T' . By [Observation 2.1\(c\)](#), let y be the unique leaf of T' adjacent to v_4 . Let $T'' := T - T_{v_2}$, and let D'', I'' and S'' be a $\gamma(T'')$ -set, an $i(T'')$ -set and an $\alpha_2(T'')$ -set, respectively. Then, we can conclude that $D := D'' \cup \{v_2\}$ is a dominating set of T implying that $\gamma(T) \leq |D| = \gamma(T'') + 1$, and $S := S'' \cup \{v_1, w\}$ is a 2-independent set of T yielding that $\alpha_2(T) \geq |S| = \alpha_2(T'') + 2$. Define

$$I := \begin{cases} I'' \cup \{v_2\}, & \text{if } v_3 \notin I'', \\ (I'' \setminus \{v_3\}) \cup \{v_2, z\}, & \text{if } v_3 \in I''. \end{cases}$$

Note that when $v_3 \in I''$, we have $\{v_4, z\} \cap I'' = \emptyset$ and thus $y \in I''$. Hence, it is easy to observe that I is an independent dominating set of T , which means that $i(T) \leq |I| = i(T'') + 1$. By the induction hypothesis, we see that

$$\gamma(T) + i(T) \leq \gamma(T'') + i(T'') + 2 \leq \frac{4}{3}\alpha_2(T'') + 2 \leq \frac{4}{3}(\alpha_2(T) - 2) + 2 = \frac{4}{3}\alpha_2(T) - \frac{2}{3} < \frac{4}{3}\alpha_2(T),$$

contradicting the assumption that $\gamma(T) + i(T) = \frac{4}{3}\alpha_2(T)$.

- Suppose v_4 is a leaf of T' . Define $T'' := T - T_{v_4} = T' - v_4$. Then $|V(T'')| = |V(T')| - 1 \geq 3$. Let D'', I'' and S'' be a $\gamma(T'')$ -set, an $i(T'')$ -set and an $\alpha_2(T'')$ -set, respectively. Then, it is straightforward to verify that $D := D'' \cup \{v_2, v_3\}$ is a dominating set of T implying that $\gamma(T) \leq |D| = \gamma(T'') + 2$, $I := I'' \cup \{v_3, v_1, w\}$ is an independent dominating set of T yielding that $i(T) \leq |I| = i(T'') + 3$, and $S := S'' \cup \{v_3, z, v_1, w\}$ is a 2-independent set of T implying that $\alpha_2(T) \geq |S| = \alpha_2(T'') + 4$. By the induction hypothesis, we derive that

$$\gamma(T) + i(T) \leq \gamma(T'') + i(T'') + 5 \leq \frac{4}{3}\alpha_2(T'') + 5 \leq \frac{4}{3}(\alpha_2(T) - 4) + 5 = \frac{4}{3}\alpha_2(T) - \frac{1}{3} < \frac{4}{3}\alpha_2(T),$$

again a contradiction.

Subcase 1.3. $k \geq 3$.

First, suppose there exists a child z of v_3 such that $3 \leq d(z) \leq k - 1$. (It is possible that $z = v_2$). Then $|L_z| \geq 2$. Let w be a vertex in L_z and let $T' := T - w$. Let D', I' and S' be a $\gamma(T')$ -set, an $i(T')$ -set and an $\alpha_k(T')$ -set, respectively. Define

$$D := \begin{cases} D', & \text{if } z \in D', \\ (D' \setminus L_z) \cup \{z\}, & \text{if } z \notin D', \end{cases}$$

and

$$I := \begin{cases} I', & \text{if } z \in I', \\ I' \cup \{w\}, & \text{if } z \notin I'. \end{cases}$$

Then, one can easily check that D is a dominating set of T implying that $\gamma(T) \leq |D| \leq \gamma(T')$ (where the second inequality holds since $L_z \setminus \{w\} \subseteq D'$ when $z \notin D'$), and I is an independent dominating set of T yielding that $i(T) \leq |I| \leq i(T') + 1$. Moreover, since $3 \leq d(z) \leq k - 1$, we notice that $S := S' \cup \{w\}$ is a k -independent set of T (no matter whether $z \in S'$ or not), which shows that $\alpha_k(T) \geq |S| = \alpha_k(T') + 1$. By the induction hypothesis, we know that

$$\begin{aligned} \gamma(T) + i(T) &\leq \gamma(T') + i(T') + 1 \leq \frac{2k}{2k-1}\alpha_k(T') + 1 \\ &\leq \frac{2k}{2k-1}(\alpha_k(T) - 1) + 1 = \frac{2k}{2k-1}\alpha_k(T) - \frac{1}{2k-1} < \frac{2k}{2k-1}\alpha_k(T). \end{aligned}$$

Next, suppose every child of v_3 has degree at most 2 or at least k . Let c_1, c_2, c_k and c_{k+1} be the numbers of children of v_3 with degree 1, 2, k and at least $k + 1$, respectively. Since $d(v_2) \geq 3$, we deduce that $c_k + c_{k+1} \geq 1$. Define $T' := T - T_{v_3}$. It is easy to see that any $\gamma(T')$ -set can be extended to a dominating set of T by adding v_3 and all its children with degree at least 2, which means that $\gamma(T) \leq \gamma(T') + 1 + c_2 + c_k + c_{k+1}$. Also, any $i(T')$ -set can be extended to an independent dominating set of T by adding all children of v_3 , which yields that $i(T) \leq i(T') + c_1 + c_2 + c_k + c_{k+1}$. Moreover, we see that any $\alpha_k(T')$ -set can be extended to a k -independent set of T by adding all children of v_3 with degree at most k and all grandchildren of v_3 , which shows that $\alpha_k(T) \geq \alpha_k(T') + c_1 + 2c_2 + kc_k + kc_{k+1}$. By the induction hypothesis, we have

$$\begin{aligned} \gamma(T) + i(T) &\leq \gamma(T') + i(T') + 1 + c_1 + 2c_2 + 2c_k + 2c_{k+1} \\ &\leq \frac{2k}{2k-1}\alpha_k(T') + 1 + c_1 + 2c_2 + 2c_k + 2c_{k+1} \\ &\leq \frac{2k}{2k-1}(\alpha_k(T) - c_1 - 2c_2 - kc_k - kc_{k+1}) + 1 + c_1 + 2c_2 + 2c_k + 2c_{k+1} \\ &= \frac{2k}{2k-1}\alpha_k(T) + \frac{2k-1-c_1-2c_2-2(k-1)^2(c_k+c_{k+1})}{2k-1}. \end{aligned} \tag{2}$$

Since $c_k + c_{k+1} \geq 1$ and $k \geq 3$, we can conclude that

$$c_1 + 2c_2 + 2(k-1)^2(c_k + c_{k+1}) \geq 2(k-1)^2 = 2k(k-2) + 2 > 2k-1.$$

Then, it follows from (2) that $\gamma(T) + i(T) < \frac{2k}{2k-1}\alpha_k(T)$.

Case 2. $d(v_2) = 2$.

We consider two subcases according to the value of k .

Subcase 2.1. $k = 1$.

Let $T' := T - T_{v_2}$, and let D' , I' and S' be a $\gamma(T')$ -set, an $i(T')$ -set and an $\alpha_1(T')$ -set, respectively. Then, it is clear that $D := D' \cup \{v_1\}$ is a dominating set of T implying that $\gamma(T) \leq |D| = \gamma(T') + 1$, $I := I' \cup \{v_1\}$ is an independent dominating set of T yielding that $i(T) \leq |I| = i(T') + 1$, and $S := S' \cup \{v_1\}$ is a 1-independent set of T implying that $\alpha_1(T) \geq |S| = \alpha_1(T') + 1$. By the induction hypothesis, we derive that

$$\gamma(T) + i(T) \leq \gamma(T') + i(T') + 2 \leq 2\alpha_1(T') + 2 \leq 2(\alpha_1(T) - 1) + 2 = 2\alpha_1(T). \quad (3)$$

Now, suppose $\gamma(T) + i(T) = 2\alpha_1(T)$. Then all inequalities in (3) should be equalities, which means that $\gamma(T') + i(T') = 2\alpha_1(T')$. By the induction hypothesis, we have $T' \in \mathcal{T}_1$. Then by Observation 2.1(b), we know that v_3 is either a support vertex or a leaf of T' .

- Suppose v_3 is a support vertex of T' . Then, we observe that T can be obtained from T' by the operation described in the definition (ii) of $\mathcal{T}_1$, and thus $T \in \mathcal{T}_1$.
- Suppose v_3 is a leaf of T' . Define $T'' := T - T_{v_3} = T' - v_3$. Since $T' \in \mathcal{T}_1$ and by Observation 2.1(a), we deduce that $|V(T')| \equiv 0 \pmod{2}$, and hence $|V(T'')| = |V(T')| - 1 \equiv 1 \pmod{2}$. Then, it follows from Observation 2.1(a) that $T'' \notin \mathcal{T}_1$. By the induction hypothesis, we have $\gamma(T'') + i(T'') < 2\alpha_1(T'')$. Let D'' , I'' and S'' be a $\gamma(T'')$ -set, an $i(T'')$ -set and an $\alpha_2(T'')$ -set, respectively. Then, it is easy to verify that $D := D'' \cup \{v_2\}$ is a dominating set of T implying that $\gamma(T) \leq |D| = \gamma(T'') + 1$, $I := I'' \cup \{v_2\}$ is an independent dominating set of T yielding that $i(T) \leq |I| = i(T'') + 1$, and $S := S'' \cup \{v_2\}$ is a 1-independent set of T implying that $\alpha_1(T) \geq |S| = \alpha_1(T'') + 1$. But then, we see that

$$\gamma(T) + i(T) \leq \gamma(T'') + i(T'') + 2 < 2\alpha_1(T'') + 2 \leq 2(\alpha_1(T) - 1) + 2 = 2\alpha_1(T),$$

which contradicts the assumption that $\gamma(T) + i(T) = 2\alpha_1(T)$.

Subcase 2.2. $k \geq 2$.

By the choice of v_2 , we conclude that all children of v_3 have degree at most 2. Let c_1 and c_2 be the numbers of children of v_3 with degree 1 and 2, respectively. Since $d(v_2) = 2$, we have $c_2 \geq 1$. Define $T' := T - T_{v_3}$.

Suppose $c_1 = 0$. Then, we derive that any $\gamma(T')$ -set can be extended to a dominating set of T by adding all children of v_3 , which shows that $\gamma(T) \leq \gamma(T') + c_2$. Also, any $i(T')$ -set can be extended to an independent dominating set of T by adding all children of v_3 , which means that $i(T) \leq i(T') + c_2$. Moreover, any $\alpha_k(T')$ -set can be extended to a k -independent set of T by adding all descendants of v_3 , which yields that $\alpha_k(T) \geq \alpha_k(T') + 2c_2$. By the induction hypothesis, we know that

$$\begin{aligned} \gamma(T) + i(T) &\leq \gamma(T') + i(T') + 2c_2 \leq \frac{2k}{2k-1}\alpha_k(T') + 2c_2 \\ &\leq \frac{2k}{2k-1}(\alpha_k(T) - 2c_2) + 2c_2 = \frac{2k}{2k-1}\alpha_k(T) - \frac{2c_2}{2k-1} < \frac{2k}{2k-1}\alpha_k(T). \end{aligned}$$

Hence, we may assume that $c_1 \geq 1$. Then, it is straightforward to check that any $\gamma(T')$ -set can be extended to a dominating set of T by adding v_3 and all its children with degree 2, which implies that $\gamma(T) \leq \gamma(T') + 1 + c_2$. Also, any $i(T')$ -set can be extended to an independent dominating set of T by adding all children of v_3 , which shows that $i(T) \leq i(T') + c_1 + c_2$. Moreover, we observe that any $\alpha_k(T')$ -set can be extended to a k -independent set of T by adding all descendants of v_3 , which means that $\alpha_k(T) \geq \alpha_k(T') + c_1 + 2c_2$. By the induction hypothesis, we deduce that

$$\begin{aligned} \gamma(T) + i(T) &\leq \gamma(T') + i(T') + 1 + c_1 + 2c_2 \\ &\leq \frac{2k}{2k-1}\alpha_k(T') + 1 + c_1 + 2c_2 \\ &\leq \frac{2k}{2k-1}(\alpha_k(T) - c_1 - 2c_2) + 1 + c_1 + 2c_2 \\ &= \frac{2k}{2k-1}\alpha_k(T) + \frac{2k-1-c_1-2c_2}{2k-1}. \end{aligned} \quad (4)$$

If $2k-1 < c_1 + 2c_2$, then by (4), we have $\gamma(T) + i(T) < \frac{2k}{2k-1}\alpha_k(T)$. Therefore, we may assume that $2k-1 \geq c_1 + 2c_2$, and thus $1 \leq c_2 \leq k-1$. We have three subcases to consider.

Subcase 2.2.1. $c_2 = k-1$.

Since $2k-1 \geq c_1 + 2c_2$ and $c_1 \geq 1$, we can conclude that $c_1 = 1$. Then $T_{v_3} \cong S_{k,k-1}$. Substituting $c_1 = 1$ and $c_2 = k-1$ into (4), we see that $\gamma(T) + i(T) \leq \frac{2k}{2k-1}\alpha_k(T)$. Now, suppose $\gamma(T) + i(T) = \frac{2k}{2k-1}\alpha_k(T)$. Then all inequalities in (4) should be equalities, which shows that $\gamma(T') + i(T') = \frac{2k}{2k-1}\alpha_k(T')$. By the induction hypothesis, we have $T' \in \mathcal{T}_k$. Since $k \geq 2$, we derive that $|V(T')| \geq 2k \geq 4$. By Observation 2.1(b), we know that v_4 is either a support vertex or a leaf of T' .

- Suppose v_4 is a support vertex of T' . Then, we notice that T can be obtained from T' by the operation described in the definition (ii) of $\mathcal{T}_k$, and hence $T \in \mathcal{T}_k$.

- Suppose v_4 is a leaf of T' . Define $T'' := T - T_{v_4} = T' - v_4$. Then $|V(T'')| = |V(T')| - 1 \geq 3$. Since $T' \in \mathcal{T}_k$ and by [Observation 2.1\(a\)](#), we deduce that $|V(T')| \equiv 0 \pmod{2k}$, and thus $|V(T'')| = |V(T')| - 1 \not\equiv 0 \pmod{2k}$. Then, it follows from [Observation 2.1\(a\)](#) that $T'' \notin \mathcal{T}_k$. By the induction hypothesis, we see that $\gamma(T'') + i(T'') < \frac{2k}{2k-1}\alpha_k(T'')$. It is clear that any $\gamma(T'')$ -set can be extended to a dominating set of T by adding v_3 and all its grandchildren, which yields that $\gamma(T) \leq \gamma(T'') + k$. Also, any $i(T'')$ -set can be extended to an independent dominating set of T by adding v_3 and all its grandchildren, which means that $i(T) \leq i(T'') + k$. Moreover, any $\alpha_k(T'')$ -set can be extended to a k -independent set of T by adding all descendants of v_3 , which shows that $\alpha_k(T) \geq \alpha_k(T'') + 2k - 1$. But then, we conclude that

$$\begin{aligned}\gamma(T) + i(T) &\leq \gamma(T'') + i(T'') + 2k < \frac{2k}{2k-1}\alpha_k(T'') + 2k \\ &\leq \frac{2k}{2k-1}[\alpha_k(T) - (2k-1)] + 2k = \frac{2k}{2k-1}\alpha_k(T),\end{aligned}$$

contradicting the assumption that $\gamma(T) + i(T) = \frac{2k}{2k-1}\alpha_k(T)$.

Subcase 2.2.2. $1 \leq c_2 \leq k-2$, and either $d(v_3) \leq k-1$ or $d(v_3) \geq k+2$.

In this subcase, we have $k \geq 3$. Let $T'' := T - T_{v_2}$, and let D'' , I'' and S'' be a $\gamma(T'')$ -set, an $i(T'')$ -set and an $\alpha_k(T'')$ -set, respectively. Then, one can easily see that $D := D'' \cup \{v_1\}$ is a dominating set of T implying that $\gamma(T) \leq |D| = \gamma(T'') + 1$, and $I := I'' \cup \{v_1\}$ is an independent dominating set of T yielding that $i(T) \leq |I| = i(T'') + 1$. Note that when $v_3 \in S''$ and $d(v_3) \geq k+2$, there must exist a child z of v_3 such that $z \neq v_2$ and $z \notin S''$. Define

$$S := \begin{cases} S'' \cup \{v_1, v_2\}, & \text{if } v_3 \notin S'' \text{ or } d(v_3) \leq k-1, \\ (S'' \setminus \{v_3\}) \cup \{v_1, v_2, z\}, & \text{if } v_3 \in S'' \text{ and } d(v_3) \geq k+2. \end{cases}$$

Since $k \geq 3$, we can derive that S is a k -independent set of T , which means that $\alpha_k(T) \geq |S| = \alpha_k(T'') + 2$. By the induction hypothesis, we know that

$$\begin{aligned}\gamma(T) + i(T) &\leq \gamma(T'') + i(T'') + 2 \leq \frac{2k}{2k-1}\alpha_k(T'') + 2 \\ &\leq \frac{2k}{2k-1}(\alpha_k(T) - 2) + 2 = \frac{2k}{2k-1}\alpha_k(T) - \frac{2}{2k-1} < \frac{2k}{2k-1}\alpha_k(T).\end{aligned}$$

Subcase 2.2.3. $1 \leq c_2 \leq k-2$ and $k \leq d(v_3) \leq k+1$.

In this subcase, we also have $k \geq 3$. Since $d(v_3) = 1 + c_1 + c_2$, we deduce that $k-1 \leq c_1 + c_2 \leq k$.

Suppose $\text{diam}(T) = 4$. Then we see that $d(v_4) = 2$; otherwise, by relabeling v_i as u_{6-i} for each $1 \leq i \leq 5$, we obtain a diametrical path $u_1u_2u_3u_4u_5$ of T with $d(u_2) \geq 3$, contradicting the choice of v_2 . It is easy to observe that the set consisting of v_5 , v_3 and all grandchildren of v_3 is both a $\gamma(T)$ -set and an $i(T)$ -set, and the set consisting of v_5 , v_4 and all descendants of v_3 is an $\alpha_k(T)$ -set. Hence, we conclude that $\gamma(T) = i(T) = 2 + c_2$ and $\alpha_k(T) = 2 + c_1 + 2c_2$. If $c_1 = 1$ and $c_2 = k-2$, then $T \cong S_{k,k-1} \in \mathcal{T}_k$. If $c_1 = 1$ and $c_2 = k-1$, then we have

$$\gamma(T) + i(T) = 2k + 2 = \frac{2k}{2k-1}(2k+1) - \frac{2}{2k-1} < \frac{2k}{2k-1}(2k+1) = \frac{2k}{2k-1}\alpha_k(T).$$

If $c_1 \geq 2$, then we derive that

$$\gamma(T) + i(T) = 4 + 2c_2 \leq 2 + c_1 + 2c_2 < \frac{2k}{2k-1}(2 + c_1 + 2c_2) = \frac{2k}{2k-1}\alpha_k(T).$$

Therefore, we may assume that $\text{diam}(T) \geq 5$. Let $C'(v_4)$ denote the set of children of v_4 that are not on any diametrical path of T containing v_{d+1} , and let $C''(v_4)$ denote the set of children of v_4 that are on a diametrical path of T containing v_{d+1} . Then by the choice of v_2 and by the above arguments, we may assume that any vertex $v \in C''(v_4)$ satisfies the same properties as v_3 :

(P1) $k \leq d(v) \leq k+1$;

(P2) there exists some $1 \leq i \leq k-2$ such that the set of children of v consists of i vertices of degree 2 and $k-1-i$ or $k-i$ vertices of degree 1.

Let r_1 and r_2 be the numbers of vertices in $C'(v_4)$ with degree 1 and at least 2, respectively. For each $1 \leq i \leq k-2$, let s_i be the number of vertices in $C''(v_4)$ with degree k whose set of children consists of i vertices of degree 2 and $k-1-i$ vertices of degree 1, and let t_i be the number of vertices in $C''(v_4)$ with degree $k+1$ whose set of children consists of i vertices of degree 2 and $k-i$ vertices of degree 1. Since $v_3 \in C''(v_4)$, we know that $\sum_{i=1}^{k-2}(s_i + t_i) = |C''(v_4)| \geq 1$.

Define $T'' := T - T_{v_4}$. Then, it is straightforward to verify that any $\gamma(T'')$ -set can be extended to a dominating set of T by adding v_4 , all its children with degree at least 2 and all its great-grandchildren, which implies that $\gamma(T) \leq \gamma(T'') + 1 + r_2 + \sum_{i=1}^{k-2}(i+1)(s_i + t_i)$. Also, any $i(T'')$ -set can be extended to an independent dominating set of T by adding all children of v_4 and all great-grandchildren of v_4 , which yields that $i(T) \leq i(T'') + r_1 + r_2 + \sum_{i=1}^{k-2}(i+1)(s_i + t_i)$. Moreover, any $\alpha_k(T'')$ -set can be extended to a k -independent set of T by adding all descendants of v_4 with degree at

most k , which shows that $\alpha_k(T) \geq \alpha_k(T'') + r_1 + 2r_2 + \sum_{i=1}^{k-2} (k+i)(s_i + t_i)$. By the induction hypothesis, we have

$$\begin{aligned} \gamma(T) + i(T) &\leq \gamma(T'') + i(T'') + 1 + r_1 + 2r_2 + 2 \sum_{i=1}^{k-2} (i+1)(s_i + t_i) \\ &\leq \frac{2k}{2k-1} \alpha_k(T'') + 1 + r_1 + 2r_2 + 2 \sum_{i=1}^{k-2} (i+1)(s_i + t_i) \\ &\leq \frac{2k}{2k-1} [\alpha_k(T) - r_1 - 2r_2 - \sum_{i=1}^{k-2} (k+i)(s_i + t_i)] + 1 + r_1 + 2r_2 + 2 \sum_{i=1}^{k-2} (i+1)(s_i + t_i) \\ &= \frac{2k}{2k-1} \alpha_k(T) + \frac{2k-1-r_1-2r_2-2 \sum_{i=1}^{k-2} (k-1)(k-1-i)(s_i + t_i)}{2k-1}. \end{aligned} \quad (5)$$

If $2k-1 < r_1 + 2r_2 + 2 \sum_{i=1}^{k-2} (k-1)(k-1-i)(s_i + t_i)$, then it follows from (5) that $\gamma(T) + i(T) < \frac{2k}{2k-1} \alpha_k(T)$. Hence, we may assume that $2k-1 \geq r_1 + 2r_2 + 2 \sum_{i=1}^{k-2} (k-1)(k-1-i)(s_i + t_i)$. Since $\sum_{i=1}^{k-2} (s_i + t_i) \geq 1$ and $k \geq 3$, we deduce that $s_i = t_i = 0$ for each $1 \leq i \leq k-3$, $s_{k-2} + t_{k-2} = 1$, $r_2 = 0$ and $0 \leq r_1 \leq 1$. This implies that v_3 is the unique vertex in $C''(v_4)$, $c_2 = k-2$ and $1 \leq c_1 \leq 2$.

First, suppose $r_1 = 0$. Then, it is clear that any $\gamma(T'')$ -set can be extended to a dominating set of T by adding v_3 and all its grandchildren, which means that $\gamma(T) \leq \gamma(T'') + k-1$. Also, any $i(T'')$ -set can be extended to an independent dominating set of T by adding v_3 and all its grandchildren, which yields that $i(T) \leq i(T'') + k-1$. Moreover, we see that any $\alpha_k(T'')$ -set can be extended to a k -independent set of T by adding all descendants of v_4 (when $c_1 = 1$) or all descendants of v_3 (when $c_1 = 2$), which shows that $\alpha_k(T) \geq \alpha_k(T'') + 2k-2$. By the induction hypothesis, we have

$$\begin{aligned} \gamma(T) + i(T) &\leq \gamma(T'') + i(T'') + 2k-2 \leq \frac{2k}{2k-1} \alpha_k(T'') + 2k-2 \\ &\leq \frac{2k}{2k-1} [\alpha_k(T) - (2k-2)] + 2k-2 = \frac{2k}{2k-1} \alpha_k(T) - \frac{2k-2}{2k-1} < \frac{2k}{2k-1} \alpha_k(T). \end{aligned}$$

Next, suppose $r_1 = 1$. Then we conclude that $|C'(v_4)| = |L_{v_4}| = 1$. Let z be the unique vertex in L_{v_4} . Substituting $r_1 = 1$, $r_2 = 0$, $s_i = t_i = 0$ ($1 \leq i \leq k-3$) and $s_{k-2} + t_{k-2} = 1$ into (5), we derive that $\gamma(T) + i(T) \leq \frac{2k}{2k-1} \alpha_k(T)$. Now, suppose $\gamma(T) + i(T) = \frac{2k}{2k-1} \alpha_k(T)$. Then all inequalities in (5) should be equalities, which means that $\gamma(T'') + i(T'') = \frac{2k}{2k-1} \alpha_k(T'')$. By the induction hypothesis, we have $T'' \in \mathcal{T}_k$. Since $k \geq 3$, we know that $|V(T'')| \geq 2k \geq 6$. By Observation 2.1(b), we deduce that v_5 is either a support vertex or a leaf of T'' .

- Suppose v_5 is a support vertex of T'' . If $c_1 = 1$, then $T_{v_4} \cong S_{k,k-1}$ and it is easy to observe that T can be obtained from T'' by the operation described in the definition (ii) of $\mathcal{T}_k$, implying that $T \in \mathcal{T}_k$. Therefore, we may assume that $c_1 = 2$. By Observation 2.1(c), let y be the unique leaf of T'' adjacent to v_5 . Define $T^* := T - T_{v_3}$. Then, we notice that any $\gamma(T^*)$ -set can be extended to a dominating set of T by adding v_3 and all its grandchildren, which yields that $\gamma(T) \leq \gamma(T^*) + k-1$. Also, any $\alpha_k(T^*)$ can be extended to a k -independent set of T by adding all descendants of v_3 , which shows that $\alpha_k(T) \geq \alpha_k(T^*) + 2k-2$. Let I^* be an $i(T^*)$ -set. If $v_4 \notin I^*$, then we let I be the set obtained from I^* by adding v_3 and all its grandchildren; otherwise, we let I be the set obtained from I^* by deleting v_4 and then adding z , v_3 and all grandchildren of v_3 . Note that when $v_4 \in I^*$, we have $\{v_5, z\} \cap I^* = \emptyset$ and thus $y \in I^*$. Hence, one can easily check that I is an independent dominating set of T , which means that $i(T) \leq |I| = i(T^*) + k-1$. By the induction hypothesis, we see that

$$\begin{aligned} \gamma(T) + i(T) &\leq \gamma(T^*) + i(T^*) + 2k-2 \leq \frac{2k}{2k-1} \alpha_k(T^*) + 2k-2 \\ &\leq \frac{2k}{2k-1} [\alpha_k(T) - (2k-2)] + 2k-2 = \frac{2k}{2k-1} \alpha_k(T) - \frac{2k-2}{2k-1} < \frac{2k}{2k-1} \alpha_k(T), \end{aligned}$$

which contradicts the assumption that $\gamma(T) + i(T) = \frac{2k}{2k-1} \alpha_k(T)$.

- Suppose v_5 is a leaf of T'' . Define $T^* := T - T_{v_5} = T'' - v_5$. Then $|V(T^*)| = |V(T'')| - 1 \geq 5$. Since $T'' \in \mathcal{T}_k$ and by Observation 2.1(a), we conclude that $|V(T'')| \equiv 0 \pmod{2k}$, and thus $|V(T^*)| = |V(T'')| - 1 \not\equiv 0 \pmod{2k}$. Then, it follows from Observation 2.1(a) that $T^* \notin \mathcal{T}_k$. By the induction hypothesis, we have $\gamma(T^*) + i(T^*) < \frac{2k}{2k-1} \alpha_k(T^*)$. First, suppose $c_1 = 1$. Then, it is straightforward to check that any $\gamma(T^*)$ -set can be extended to a dominating set of T by adding v_4 and all children of v_3 , which implies that $\gamma(T) \leq \gamma(T^*) + k$. Also, any $i(T^*)$ -set can be extended to an independent dominating set of T by adding v_4 and all children of v_3 , which shows that $i(T) \leq i(T^*) + k$. Moreover, any $\alpha_k(T^*)$ -set can be extended to a k -independent set of T by adding all descendants of v_4 , which yields that $\alpha_k(T) \geq \alpha_k(T^*) + 2k-1$. But then, we derive that

$$\gamma(T) + i(T) \leq \gamma(T^*) + i(T^*) + 2k < \frac{2k}{2k-1} \alpha_k(T^*) + 2k \leq \frac{2k}{2k-1} [\alpha_k(T) - (2k-1)] + 2k = \frac{2k}{2k-1} \alpha_k(T),$$

a contradiction.

Next, suppose $c_1 = 2$. It is clear that any $\gamma(T^*)$ -set can be extended to a dominating set of T by adding v_4 , v_3 and all grandchildren of v_3 , which means that $\gamma(T) \leq \gamma(T^*) + k$. Also, any $i(T^*)$ -set can be extended to an independent dominating set of T by adding v_4 and all children of v_3 , which implies that $i(T) \leq i(T^*) + k + 1$. Moreover, we notice that any $\alpha_k(T^*)$ -set can be extended to a k -independent set of T by adding v_4 , z and all descendants of v_3 , which shows that $\alpha_k(T) \geq \alpha_k(T^*) + 2k$. But now, we deduce that

$$\begin{aligned} \gamma(T) + i(T) &\leq \gamma(T^*) + i(T^*) + 2k + 1 < \frac{2k}{2k-1} \alpha_k(T^*) + 2k + 1 \\ &\leq \frac{2k}{2k-1} (\alpha_k(T) - 2k) + 2k + 1 = \frac{2k}{2k-1} \alpha_k(T) - \frac{1}{2k-1} < \frac{2k}{2k-1} \alpha_k(T), \end{aligned}$$

again a contradiction.

This completes the proof of [Theorem 2.2](#). ■

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Data availability

No data was used for the research described in the article.

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